

PORI TAHKOLUOTO HARBOUR, Finland

WMO: 027510

Lat: 61.630N

Lon: 21.376E

Elev: 5

StdP: 101.27

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-14.7	-19.7	0.7	-16.7	-16.7	0.9	-13.9	19.2	0.7	17.6	0.6	4.3	110	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	23.5	19.5	21.8	18.5	20.4	17.8	20.6	22.7	19.2	21.1	18.1	20.0	4.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.4	22.1	18.4	13.3	20.8	17.2	12.3	19.5	59.4	22.8	54.8	21.1	51.1	20.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.7	14.2	12.9	DB	-19.2	26.4	3.7	2.5	-21.9	28.2	-24.0	29.7	-26.1	31.1	-28.9	33.0	
			WB	-18.7	20.7	4.4	1.6	-21.8	21.9	-24.4	22.9	-26.9	23.8	-30.1	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-3.9	-3.2	-1.1	2.9	8.8	12.7	16.7	16.4	13.0	6.9	3.2	-0.2
	DBStd	8.04	5.13	5.21	3.33	2.51	3.63	3.00	3.09	2.43	2.54	2.92	3.74	4.94
	HDD10.0	2062	432	371	343	214	67	7	0	0	7	100	205	315
	HDD18.3	4509	691	604	602	463	296	173	71	68	161	353	455	573
	CDD10.0	627	0	0	0	2	30	87	208	200	96	4	0	0
	CDD18.3	33	0	0	0	0	1	3	20	9	0	0	0	0
	CDH23.3	59	0	0	0	0	2	5	30	22	0	0	0	0
	CDH26.7	5	0	0	0	0	0	1	3	2	0	0	0	0
Wind	WSAvg	6.5	7.0	6.9	6.7	5.7	5.8	5.8	5.5	6.2	6.4	7.0	7.3	8.3
Precipitation	PrecAvg	630	51	37	33	32	38	56	68	71	58	69	60	58
	PrecMax	818	100	78	58	61	85	108	134	137	128	159	139	108
	PrecMin	514	7	2	5	1	5	18	2	25	12	7	9	7
	PrecStd	77	23	20	13	16	20	23	33	32	35	35	28	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	3.6	6.1	14.9	22.4	23.6	26.0	26.1	20.1	13.7	10.0	7.1
		MCWB	3.2	2.5	2.9	9.0	14.1	17.0	20.8	20.8	16.8	12.1	9.0	6.3
	2%	DB	3.4	2.8	4.0	9.6	18.0	20.8	23.9	22.8	17.8	12.3	9.0	6.1
		MCWB	2.6	1.9	2.7	6.5	14.3	17.0	20.2	19.5	15.6	10.9	8.4	5.1
	5%	DB	2.7	2.2	3.2	7.0	16.0	18.6	22.4	21.2	16.9	11.5	8.0	5.2
		MCWB	2.1	1.6	2.2	5.1	13.4	15.9	19.5	18.3	15.2	10.1	7.1	4.2
	10%	DB	2.0	1.7	2.6	5.7	13.8	16.7	20.9	19.5	16.1	10.5	7.2	4.5
		MCWB	1.4	1.2	1.7	4.3	11.2	14.4	18.7	17.0	14.5	9.3	6.5	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	2.7	3.8	10.1	16.1	18.8	22.1	22.0	17.1	12.4	9.2	6.6
		MCDB	3.8	3.2	5.0	14.1	19.2	21.6	24.9	24.6	19.3	13.3	9.7	7.0
	2%	WB	2.9	2.1	2.9	7.0	14.8	17.3	21.1	20.1	16.1	11.3	8.6	5.2
		MCDB	3.2	2.5	3.7	9.4	18.0	20.1	23.3	22.2	17.4	12.1	9.0	5.9
	5%	WB	2.1	1.7	2.4	5.4	13.3	16.1	20.0	18.7	15.4	10.5	7.3	4.3
		MCDB	2.5	2.0	3.1	6.5	16.1	18.5	21.8	20.6	16.5	11.1	7.8	4.9
	10%	WB	1.5	1.3	1.9	4.5	11.3	14.7	18.9	17.6	14.8	9.6	6.5	3.8
		MCDB	1.9	1.6	2.5	5.5	13.5	16.4	20.6	19.1	15.8	10.3	7.0	4.3
Mean Daily Temperature Range	5% DB	MDBR	4.1	3.6	3.9	3.3	3.9	3.6	3.6	3.3	3.2	2.9	2.6	3.0
		MCDBR	2.7	2.4	3.3	5.7	6.8	6.7	5.7	5.2	4.2	2.7	2.5	2.7
	5% WB	MCWBR	2.4	2.1	2.6	3.7	3.9	3.7	3.3	3.3	2.9	2.7	2.5	2.8
		MCDWR	2.5	2.4	3.2	5.1	6.5	6.1	4.7	4.1	3.5	2.6	2.3	2.6
Clear-Sky Solar Irradiance	taub	taub	0.264	0.308	0.326	0.366	0.364	0.356	0.382	0.369	0.344	0.327	0.283	0.350
		taud	2.137	2.239	2.313	2.329	2.420	2.479	2.431	2.456	2.506	2.394	2.186	2.519
	Ebn at Noon	Ebn at Noon	436	654	787	823	858	872	834	816	772	635	404	277
		Edh at Noon	36	65	85	102	101	97	100	90	72	56	35	23
All-Sky Solar Radiation	RadAvg	0.18	0.74	2.27	3.92	5.55	6.11	5.61	4.36	2.58	1.05	0.27	0.11	
	RadStd	0.02	0.14	0.26	0.40	0.46	0.35	0.47	0.38	0.26	0.18	0.05	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page